FareShare

Campus Ride-Pooling & Fair Distance-Split Platform

MVP MILESTONE SUBMISSION

Event: **coNStruCT 2.0 Hackathon (NST × The Product Folks)** Team: **Team EnderMen (5 Builders)** Build: **Production Live (Vite 6 / React 19)**

75%	< 50ms	4 Hubs	1-Click
SOLO FARE REDUCTION	MATCHING LATENCY	TRANSIT CORRIDORS	DIRECT UPI SPLIT

01. Core Interactive Capabilities (What We Have Built)

FareShare is a fully functional web platform designed specifically for university students facing exorbitant solo cab fares on common transit routes. Rather than submitting static concepts, Team EnderMen built and deployed an operational end-to-end system with the following architectural pillars:

Verified Campus Safe Circles & Capacity Filters

Enforces verified university email domains for riders and hosts. Features a dedicated Girls-Only safe circle toggle for verified female student pooling, paired with luggage capacity matching (Backpack, 1 Medium Trolley, Heavy Boot Space) to prevent overloaded vehicles.

Bilingual Conversational Intent Parser (Voice & Text)

Integrated directly with the browser Web Speech API and an in-client algorithmic parser. Students speak or type informal English, Hindi, or Hinglish intent (e.g., "kal shaam IGI T3 jana hai") to extract destination, departure window, and urgency without cloud API latency.

Real-Time Multi-Modal Market Benchmarks

Every ride pool card embeds live deep links to Uber Go, Ola Mini, and the Delhi Metro (DMRC) fare engine. This allows students to verify prevailing solo rates in real time, establishing transparent price parity and confirming the 75% savings before committing.

1-Click Universal UPI & Ephemeral Privacy Rooms

Coordinates settlement via pre-populated NPCI Universal UPI deep-links (Google Pay, PhonePe, Paytm) carrying the exact calculated fare split. Ride chatrooms and real-time pickup coordinates auto-purge post-trip to ensure student data privacy.

02. Real-Time Economic Impact (Solo Cab vs. FareShare 4-Way Pool)

CAMPUS TRANSIT CORRIDOR	SOLO UBER GO	SOLO OLA MINI	DELHI METRO (DMRC)	FARESHARE (4-WAY)	STUDENT SAVINGS
Campus ⇄ IGI Airport T3 (44 KM • ~65 min)	₹1,400	₹1,350	₹60 (80 min, 2 transfers)	**₹350 / rider**	Save ₹1,050 (75%)
Campus ⇄ NDLS Railway Station (31 KM • ~50 min)	₹900	₹880	₹50 (65 min, Yellow Line)	**₹225 / rider**	Save ₹675 (75%)
Campus ⇄ Anand Vihar ISBT (28 KM • ~45 min)	₹800	₹780	₹50 (60 min, Blue Line)	**₹200 / rider**	Save ₹600 (75%)
Campus ⇄ Cyber City / Metro Hub (14 KM • ~25 min)	₹450	₹430	₹30 (40 min, Rapid Line)	**₹113 / rider**	Save ₹337 (75%)

Information Architecture & Wireframes

System Mechanics, Low-Fidelity Layout Hierarchy & UI/UX Principles

STRUCTURAL BLUEPRINTS

SYSTEM MECHANICS & DATA FLOW PIPELINE

STAGE 01	STAGE 02	STAGE 03	STAGE 04	STAGE 05
Intent Capture	**Filter Engine**	**Cost Division**	**Seat Locking**	**Settlement**
Voice / Hinglish Omnibox query parsed client-side	Gender safety circles & boot capacity evaluation	Proportional distance split against market rate	Optimistic inventory decrement & OTP generation	NPCI UPI deep link & 4h ephemeral session purge

02. Low-Fidelity Screen Schematics (Layout Structure & Interaction Flow)

Screen 01: Omnibox & Corridor Discovery

LOW-FI WIREFRAME

```
[ SEARCH_INPUT: "Campus Gate 1 → IGI Airport T3" | MIC_TRIGGER ]
[ ROUTE: Airport T3 ]  [ ROUTE: NDLS ]  [ ROUTE: Cyber City ]
[ TOGGLE: Girls-Only Circle ]  [ SELECT: Boot Luggage ]
```

Structure: Replaces deep multi-level forms with a single unified conversational input container, active corridor pill selectors, and primary safety toggles.

Screen 02: Pool Feed & Market Parity Card

LOW-FI WIREFRAME

```
[ HOST_META: Avatar • Name • Rating 4.9 • .edu ]  [ SEATS_STATUS: 2 Left ]
[ ROUTE_VECTOR: Origin ——————→ Destination (44 KM) ]
[ BENCHMARKS: Solo Uber ₹1400 | Solo Ola ₹1350 | Metro ₹60 ]
[ FARE: ₹350 (-75%) ]  [ ACTION: Request Pool ]
```

Structure: Embeds market solo pricing directly into the pool card container, giving immediate psychological price confirmation prior to interaction.

Screen 03: Reservation & Digital Boarding Pass

LOW-FI WIREFRAME

```
[ PASS_ID: #FS-2026-89 ]  [ STATUS: CONFIRMED ]
[ PICKUP_POINT: Campus Gate 1 | DEPARTURE: 06:30 PM ]
[ PASSENGER_MANIFEST: Host (1) + Co-Riders (2/3 Assigned) ]
[ VERIFICATION_OTP: 8924 ]  [ LUGGAGE_CHECK: 3/4 Bags ]
```

Structure: High-density confirmation sheet displaying critical pickup coordinates, OTP for vehicle boarding, and peer manifest without clutter.

Screen 04: Universal UPI & Ephemeral Room

LOW-FI WIREFRAME

```
[ SPLIT_DUE: ₹350.00 ]  [ PURGE_TIMER: 03:59:40 ]
[ CALCULATION: Total Cab ₹1,400 ÷ 4 Verified Riders = ₹350 ]
[ GPay Intent → ]  [ PhonePe Intent → ]
[ PRIVACY_POLICY: Coordination history & numbers self-destruct post-trip ]
```

Structure: Uses native intent links to trigger third-party UPI applications with pre-filled payment fields, followed by an automated session expiry protocol.

03. UI/UX Design System Principles

Reduced Cognitive Load

Students planning transit between classes need immediate utility. The single conversational omnibox eliminates nested configuration funnels in favor of natural language inputs.

Structural Safety by Default

Girls-Only Safe Circles and .edu institutional email requirements are built into the primary information architecture, rather than hidden behind nested account settings.

Touch-First Ergonomics

Optimized for one-handed mobile navigation with 48px target heights, swipeable drawer containers, high-contrast typography, and zero-login parity cross-checks.

User Journey, Validation & Roadmap

Operational Flow, Student Walkthrough Insights & Production Expansion

EXECUTION STRATEGY

04. End-to-End User Journey (Current Working Flow)

STEP 01

Discovery & Filter Customization

Riders enter their schedule via conversational voice or corridor selectors. They toggle Girls-Only mode if preferred and declare luggage size. The matching engine filters active pools in real time.

STEP 02

Multi-Modal Solo Rate Cross-Check

Before initiating a request, students click comparative link pills to Uber, Ola, or DMRC Metro to inspect live market rates without authentication, validating the cost advantage.

STEP 03

Instant Booking & Boarding Pass

Clicking "Request to Pool" decrements open capacity in state, locks the seat against concurrent users, and produces an active digital boarding pass containing the 4-digit pickup OTP.

STEP 04

Ride Hosting & Feed Publishing

If no existing pool aligns with a commuter's schedule, they click "+ Post Ride Pool". The system calculates fair proportional splits based on corridor distance and publishes it to the campus feed.

05. User Testing & Iterative Feedback

Direct usability walkthroughs with university students commuting between Greater Noida/NCR and regional hubs revealed three critical friction points that guided the current architecture:

Safety Concerns in Open Pools

Female students expressed reluctance to pool with unverified strangers. In response, we implemented verified .edu institutional authentication and explicit Girls-Only circles.

Boot Space Overcrowding

Early tests revealed airport travellers often brought multiple large suitcases. We introduced luggage volume tiering (Backpack, 1 Trolley, Large Boot) to ensure bags fit comfortably.

Post-Trip Payment Friction

Testers disliked calculating splits or chasing peers for manual transfers. We replaced manual settlements with 1-click NPCI universal UPI deep links containing pre-filled amounts.

06. Production Roadmap: Scaling from MVP to Pan-India

Expansion Vision: Preset Corridors to National University Transit Mesh

- **Google Maps Platform Routing:** Transition from preset routes to arbitrary pin-to-pin navigation utilizing the *Google Maps Distance Matrix* and *Directions API* for algorithmic intermediate drop-off calculations.
- **Inter-College Regional Networks:** Expanding single-campus deployments into shared clusters connecting proximate universities and polytechnics along common transit routes.
- **Commercial Dispatch Integration:** Official webhooks into commercial mobility dispatch APIs (Uber/Ola) for automated vehicle reservation directly from confirmed pool rooms.

LIVE PRODUCTION REPLIT APP

`https://fareshare--pvats2112.replit.app`

OPEN-SOURCE GITHUB REPOSITORY

`https://github.com/Prachivats-21/fareshare`